30/07/2025

Code-A

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18,
Udyog Vihar, Gurugram, Haryana - 122015, *Ph.*+91-1244168300

MM : 720 NCERT Booster Test Series for NEET-2026_RM(P1)_NBTS-01A **Time : 180 Min.**

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy & Power, System of Particles & Rotational Motion
Chemistry: Some Basic Concepts of Chemistry, Structure of Atom, Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Botany: Cell : The Unit of Life, Cell Cycle & Cell Division, The Living World, Biological Classification
Zoology: Structural Organisation in Animals– Animal Tissues, Biomolecules, Breathing & Exchange of Gases, Body Fluids & Circulation

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use whitener/liquid or any other rubbing material on the Answer sheet.

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PHYSICS

1. An object is thrown vertically upwards. It has velocity of 20 m/s when it reaches half of its maximum height. If g = 10 m/s^2, then the maximum height reached by the object will

(1) 80 m

(2) 60 m

(3) 40 m

(4) 100 m

2. A particle starts moving with an acceleration of 4 m/s^2. Distance travelled by it in 3rd second of motion is

(1) 40 m

(2) 20 m

(3) 10 m

(4) 30 m

3. Given $y = \cos(2x) - 2\sin x$, then value of $\frac{dy}{dx}$ is

(1) $-(\sin 2x + 2\cos x)$

(2) $-(\sin x - \cos x)$

(3) $-2\cos x\,(2\sin x + 1)$

(4) $(-\cos 2x + \sin x)$

4. A train moving at 10 m s^{-1} takes 12 second to cross a bridge of length 80 m. The length of the train is

(1) 60 m

(2) 40 m

(3) 20 m

(4) 10 m

5. A particle starts from origin, moves along *x*-axis to the point (5 m, 0) and then returns along the same line to the point (–10 m, 0). The displacement of the particle during the trip is

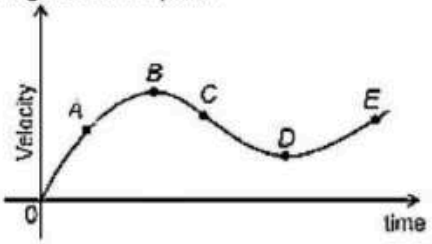
(1) 10 m

(2) –10 m

(3) 15 m

(4) –20 m

6. The velocity-time graph of a moving particle is shown in graph. The instantaneous acceleration of the particle is negative at the point



(1) *A*

(2) *D*

(3) *E*

(4) *C*

7. A : It is not possible to have constant velocity and variable acceleration.
R : Accelerated body cannot have constant velocity.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

8. A swimmer swims with a speed 5 m/s w.r.t. river. Minimum time, that swimmer would take, to cross the river of width 300 m will be

(1) 60 s

(2) 30 s

(3) 10 s

(4) 5 s

9. Two projectiles are fired in the same direction with different initial speeds. If the ratio of their initial speeds is 2 : 3, then the ratio of their respective horizontal ranges will be

(1) 2 : 3

(2) 3 : 2

(3) 4 : 9

(4) 9 : 4

10. A particle is moving in a circle with constant speed of 7 m/s, then the centripetal acceleration of particle is (If particle covers 44 meter to complete one circular round)

(1) 2 m/s^2

(2) 7 m/s^2

(3) 3 m/s^2

(4) 5 m/s^2

11. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion (A) : For unit vectors $\hat{i}$ and $\hat{j}$, $\left(\hat{i}+\hat{j}\right)=1$

Reason (R) : Sum of two unit vectors is also a unit vector.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

12. A body is projected from the top of a tower of height 20 m with speed 20 m/s along horizontal (positive *x*-direction). The velocity vector with which it will hit the ground will be (g = 10 m/s^2, Take vertically upward direction as positive *y*-axis)

(1) $\left(20\hat{i}-10\hat{j}\right)\ \mathrm{m/s}$

(2) $\left(20\hat{i}+10\hat{j}\right)\ \mathrm{m/s}$

(3) $\left(20\hat{i}-20\hat{j}\right)\ \mathrm{m/s}$

(4) $\left(20\hat{i}+20\hat{j}\right)\ \mathrm{m/s}$

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13. Choose the correct statement about a particle undergoing uniform circular motion

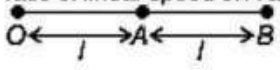
(1) Velocity remains constant

(2) Speed remains constant

(3) Acceleration remains constant

(4) Acceleration is zero

14. Two identical particles are joined together by a thread as shown in figure. Both particles are moving in a circular path about *O* in a horizontal plane with same angular velocity the ratio of linear speed of *A* and *B* is



(1) 1 : 2

(2) 1 : 3

(3) 2 : 3

(4) 3 : 5

15. The diameter of a spherical body, when measured with vernier callipers yielded the following values: 4.33 cm, 4.32 cm, 4.38 cm, 4.31 cm and 4.32 cm. The mean diameter to appropriate significant figures is

(1) 4.3 cm

(2) 4.33 cm

(3) 4.332 cm

(4) 4.4 cm

16. If the force (F) is given by $F = at + bt^2$ (where t is time), then dimensional formula of ab will be

(1) $[M^2L^2T^{-3}]$

(2) $[ML^2T^{-3}]$

(3) $[M^2L^2T^{-7}]$

(4) $[MLT^{-6}]$

17. $[ML^2T^{-3}]$ represents the dimensional formula of

(1) Work

(2) Pressure

(3) Power

(4) Force

18. If $x = ab$, and Δa represent absolute error in measurement of a and Δb represents absolute error in measurement of b, then the absolute error in x (Δx) equals

(1) $\frac{a\Delta a+b\Delta b}{a+b-ab}$

(2) $\frac{\Delta a}{a}+\frac{\Delta b}{b}$

(3) $a\Delta a + b\Delta b$

(4) $b\Delta a + a\Delta b$

19. Consider the diameter of a spherical object being measured with the help of a Vernier callipers. Suppose its 10 Vernier Scale Divisions (V.S.D.) are equal to its 9 Main Scale Divisions (M.S.D.). The least division in the M.S. is 0.1 cm and the zero of V.S. is at $x = 0.1$ cm when the jaws of Vernier callipers are closed.
If the main scale reading for the diameter is $M = 5$ cm and the number of coinciding vernier division is 8, the measured diameter after zero error correction, is

(1) 4.98 cm

(2) 5.00 cm

(3) 5.18 cm

(4) 5.08 cm

20. The number of significant figures in measured value 0.0602 g is

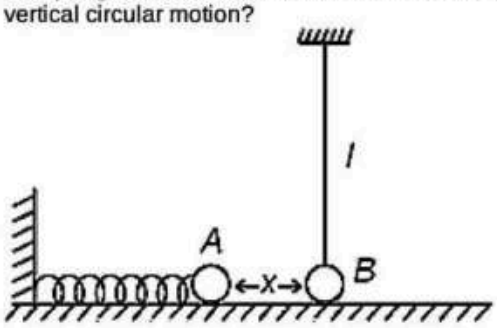
(1) 4

(2) 5

(3) 3

(4) 2

21. A small ball 'A' of mass 'm' is pressed against a spring of force constant K and then released which further collides elastically with another identical ball 'B' connected with an inextensible thread as shown in the figure. If compression in the spring is x, then for what value of x ball 'B' completes its vertical circular motion?



(1) $\sqrt{\frac{mgl}{K}}$

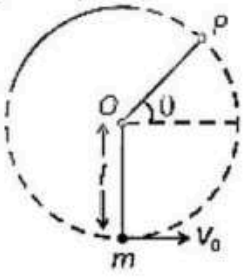
(2) $\sqrt{\frac{5mgl}{K}}$

(3) $\sqrt{\frac{mgl}{2K}}$

(4) $\sqrt{\frac{4mgl}{K}}$

22. A bob of heavy mass m is suspended by a light string of length l. The bob is given a horizontal velocity v_0 as shown in figure. If the string gets slack at some point P making an angle θ from the horizontal, the ratio of the speed v of the bob at point P to its initial speed v_0 is:

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(1) $\left(\frac{\cos\theta}{2+3\sin\theta}\right)^{\frac{1}{2}}$

(2) $\left(\frac{\sin\theta}{2+3\sin\theta}\right)^{\frac{1}{2}}$

(3) $(\sin\theta)^{\frac{1}{2}}$

(4) $\left(\frac{1}{2+3\sin\theta}\right)^{\frac{1}{2}}$

23. A body of mass 10 kg moves along a smooth inclined plane ($\theta = 30^o$) with constant velocity through a distance of 10 m. Calculate the work done by applied external force to move the body through desired distance of 10 m.

(1) 100 J

(2) 150 J

(3) 500 J

(4) 400 J

24. The potential energy for a conservative force system is given by $U = bx^2 - cx$, where b and c are constants. The equilibrium position for the system is

(1) $\frac{2b}{c}$

(2) $\frac{3b}{c^2}$

(3) $\frac{c}{2b}$

(4) $2bc$

25. Match List- I with List-II and choose the correct option (where symbols have their usual meaning).

	List-I		List-II
A	Work done by all the forces	p.	$\frac{\text{Area under } P-t \text{ graph}}{\text{Total time interval}}$
B	Work done by conservative forces	q.	Change in kinetic energy
C	Average power	r.	Negative of change in potential energy
D	Instantaneous power	s.	Slope of W-t graph

(1) A→ r, B → q, C → s, D → p

(2) A→ q, B → r, C → s, D → p

(3) A→ q, B → r, C → p, D → s

(4) A→ r, B → q, C → p, D → s

26. A body is initially at rest. It undergoes one dimensional motion with acceleration proportional to time i.e. $a \propto t$. The power delivered to it at time t is proportional to

(1) t

(2) t^2

(3) t^3

(4) t^4

27. Identify the set of correct statements
(A) A force $F(x)$ is conservative if it can be derived from a scalar quantity $V(x)$, with the help of the relation given as $\Delta V = -F(x)\Delta x$.
(B) The work done by the conservative force depends only on the initial and final positions.
(C) Spring force is a type of conservative force.
(D) Work done by static friction on a body may be zero.
Choose the correct answer from the options given below

(1) Only (A), (B) and (C)

(2) Only (A), (B) and (D)

(3) Only (B), (C) and (D)

(4) All (A), (B), (C) and (D)

28. A thin uniform rod of mass m and length l is swinging freely about a horizontal axis passing through its one end. When it is at bottom-most position as shown in figure its angular velocity is ω. If centre of mass of the rod can rise to a maximum height $\frac{l}{4}$ from this position, then ω is



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(1) $\sqrt{\frac{3g}{l}}$

(2) $\sqrt{\frac{3}{2}\frac{g}{l}}$

(3) $\sqrt{\frac{2}{3}\frac{g}{l}}$

(4) $\frac{1}{2}\sqrt{\frac{3g}{l}}$

29. A rigid body is rotating about a fixed axis with angular velocity ω. Its moment of inertia is I and kinetic energy is K. Its angular momentum about the same axis is

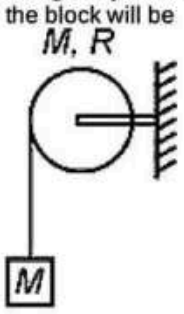
(1) $\sqrt{2KI}$

(2) $\frac{2K}{\omega}$

(3) $2K\omega$

(4) Both (1) and (2)

30. A disc of mass M and radius R is free to rotate about its centre as shown in figure. A string is tightly wrapped over its rim and block of mass M is attached to the free end of the string. If system is released from rest, then acceleration of the block will be



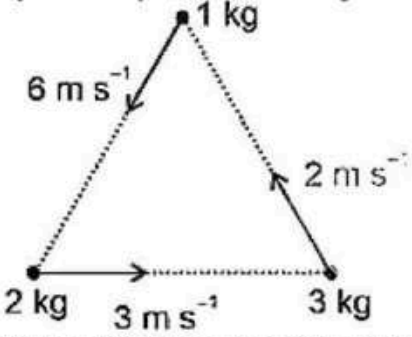
(1) $\frac{g}{3}$

(2) $\frac{3g}{4}$

(3) $\frac{g}{4}$

(4) $\frac{2g}{3}$

31. Three particles of masses 1 kg, 2 kg and 3 kg are at the vertices of an equilateral triangle moving at a speed of 6 m s^{-1}, 3 m s^{-1} and 2 m s^{-1} respectively. Each particle maintains a direction towards the particle at the next vertex symmetrically as shown in the figure.



Velocity of centre of mass of the system at this instant is

(1) $\sqrt{63}$ m s^{-1}

(2) $\sqrt{128}$ m s^{-1}

(3) 8 m s^{-1}

(4) Zero

32. Two masses 6 kg and 4 kg are connected by massless flexible and inextensible string passing over massless and frictionless pulley. The magnitude of acceleration of centre of mass is

(g = 10 ms^{-2})

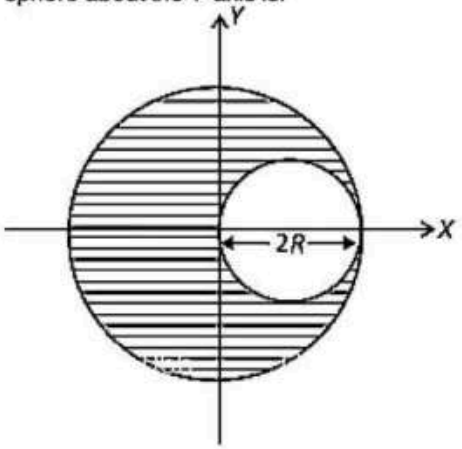
(1) 0.4 ms^{-2}

(2) 2 ms^{-2}

(3) 50 ms^{-2}

(4) Zero

33. A sphere of radius R is cut from a larger solid sphere of radius $2R$ as shown in the figure. The ratio of the moment of inertia of the smaller sphere to that of the rest part of the sphere about the Y-axis is:



(1) $\frac{7}{57}$

(2) $\frac{7}{64}$

(3) $\frac{7}{8}$

(4) $\frac{7}{40}$

34. The moment of inertia of a uniform semicircular ring of mass M and radius R about an axis perpendicular to plane of the ring and passing through its centre is

(1) MR^2

(2) $\frac{MR^2}{2}$

(3) $2MR^2$

(4) $\frac{3}{4}MR^2$

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35. A particle is moving in a circle with uniform speed. Its angular momentum about centre of the circle is constant due to

(1) Constant non-zero torque

(2) Constant force

(3) Constant linear momentum

(4) Zero torque

36. Two masses m_1 and m_2 are d distance apart, where $m_1 > m_2$. The centre of mass of the system will

(1) Lie at a distance $\frac{d}{2}$ from m_1

(2) Be closer to m_2

(3) Be closer to m_1

(4) Lie at distance d from m_2

37. For a given surface, the normal force and friction force are inclined

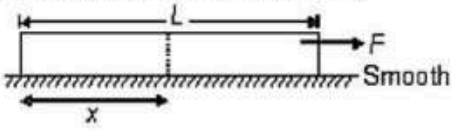
(1) At 90° to each other

(2) At zero degree to each other

(3) Both at zero degree to the surface

(4) Both at 90° to the surface

38. Tension T in the string of mass M at the distance 'x' as shown in figure from left is given by



(1) $T = \frac{Mx}{L}$

(2) $T = \frac{FL}{x}$

(3) $T = \frac{F(L-x)}{L}$

(4) $T = \frac{F(L-x)}{x}$

39. Velocity of a particle of mass 2 kg varies with time according to the equation $v = \left(2t\hat{i} + 4\hat{j}\right)$ m/s. Here t is in second. The impulse imparted to the particle in time interval from $t = 0$, $t = 2$ s is

(1) 8 Ns

(2) 10 Ns

(3) 6 Ns

(4) 4 Ns

40. Calculate the maximum acceleration of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is 0.25 ($g = 10\ m\ s^{-2}$)

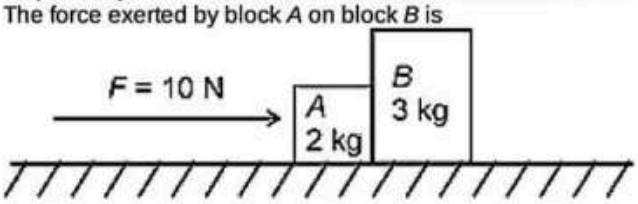
(1) $25\ m\ s^{-2}$

(2) $2.5\ m\ s^{-2}$

(3) $1.5\ m\ s^{-2}$

(4) $0.75\ m\ s^{-2}$

41. A horizontal force 10 N is applied to a block A as shown in figure. The mass of blocks A and B are 2 kg and 3 kg respectively. The blocks slide over a frictionless surface. The force exerted by block A on block B is



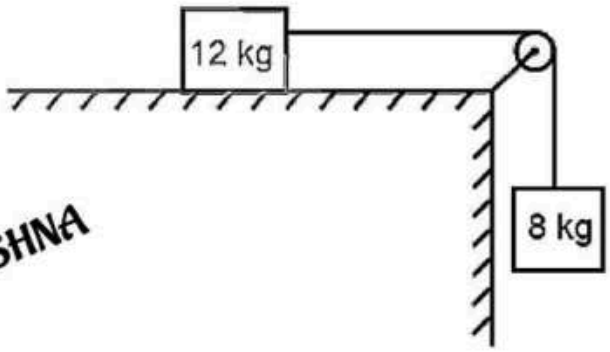
(1) Zero

(2) 4 N

(3) 6 N

(4) 10 N

42. A block of mass 12 kg is kept on a rough horizontal surface. It is connected to a string, which passes over a pulley. When a weight of 8 kg is attached to the other end of the string, then the block just begins to move, then coefficient of static friction is equal to



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(1) $\frac{3}{2}$

(2) $\frac{2}{3}$

(3) $\frac{1}{2}$

(4) $\frac{1}{3}$

43. A machine gun fires 5 bullets per second and the mass of each bullet is 20 g. If 100 m/s is the speed of each bullet, then the force exerted on the machine gun is

(1) 10 N

(2) 20 N

(3) 8 N

(4) 15 N

44. One end of a string of length *l* is connected to a particle of mass *m* and the other end is connected to a small peg on a smooth horizontal table. If the particle moves in a horizontal circle with speed *v*, then the magnitude of net force on the particle is

(1) zero

(2) $\frac{mv^2}{l}$

(3) $\sqrt{(mg)^2+\left(\frac{mv^2}{l}\right)^2}$

(4) *mg*

45. A diwali rocket with a lift-off mass 100 g is blasted upwards with an initial acceleration of 20 m/s^2. The initial thrust(force) of the blast is

(1) 2 N

(2) 3 N

(3) 2.5 N

(4) 4 N

CHEMISTRY

46. By burning 10 g of hydrogen with 90 g of oxygen, the mass of water formed is

(1) 40 g

(2) 80 g

(3) 90 g

(4) 100 g

47. Number of S-atoms present in 0.5 g molecules of S_8 is

(1) 1.98×10^{23} atoms

(2) 2.14×10^{23} atoms

(3) 2.41×10^{24} atoms

(4) 1.25×10^{24} atoms

48. The mass percentage of oxygen in H_2SO_4 is

(1) 15.23

(2) 65.31

(3) 17.15

(4) 23.24

49. If mass percentage of carbon, oxygen and hydrogen in an organic compound are 26.67%, 71.11% and 2.22% respectively then empirical formula of the organic compound will be

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(1) CH_2O

(2) CHO_2

(3) C_2HO_2

(4) CHO

50. Identify the incorrect match.

	Multiple	Prefix
(1)	10^{-15}	Femto
(2)	10^{-12}	Pico
(3)	10^{15}	Tera
(4)	10^{2}	Hecto

(1) (1)

(2) (2)

(3) (3)

(4) (4)

51. Given below are two statements:
Statement I : Molecular formula mass of benzene is six times of its empirical formula mass.
Statement II : An empirical formula represents the simplest whole number ratio of various atoms present in a compound.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) Statement I is incorrect but statement II is correct.

(2) Statement I is correct but statement II is incorrect.

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect.

52. Volume of O_2 required at STP to react completely with 180 g of ethane is

(1) 470.4 L

(2) 448 L

(3) 134.4 L

(4) 224 L

53. Mass percent of glucose in an aqueous solution is 18%. The molality of glucose solution is

(1) 1.22 m

(2) 1.86 m

(3) 2.21 m

(4) 0.75 m

54. If 3.01×10^{22} molecules of urea are present in 50 mL of an aqueous solution then the molarity of the solution will be

(1) 0.5 M

(2) 1.5 M

(3) 1 M

(4) 2.5 M

55. 200 ml of 3 M NaOH solution is diluted by 400 ml water. The molarity of the resulting solution is

(1) 1 M

(2) 2 M

(3) 1.5 M

(4) 2.25 M

56. Consider the following statements about Dalton's atomic theory
(a) All atoms of a given element have identical properties.
(b) Atoms are neither created nor destroyed in a chemical reaction.
(c) Matter consists of indivisible atoms.
The correct statements are

(1) (a), (b) and (c)

(2) (a) and (b) only

(3) (a) and (c) only

(4) (b) and (c) only

57. The mole fraction of glucose in 0.1 molal aqueous solution of glucose will be

(1) 0.017

(2) 0.0017

(3) 0.17

(4) 0.25

58. Given below are two statements one is labelled as assertion (A) and other is labelled as reason (R).
Assertion (A) : The second ionization enthalpy will be higher than the first ionization enthalpy of an atom.
Reason (R) : It is more difficult to remove an electron from a positively charged ion than from a neutral atom due to increase in effective nuclear charge.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

59. An atom with which of the following electronic configuration has the highest ionization enthalpy?

(1) $1s^2 2s^2 2p^1$

(2) $1s^2 2s^2$

(3) $1s^2 2s^2 2p^3$

(4) $1s^2 2s^2 2p^4$

60. Match column I with column II and choose the correct answer.

	Column I		Column II
a.	Palladium	(i)	Lanthanoid
b.	Chlorine	(ii)	Transition metal
c.	Thorium	(iii)	Halogen
d.	Samarium	(iv)	Actinoid

Choose the correct answer from the options given below.

(1) a-(i), b-(iii), c-(iv), d-(ii)

(2) a-(iv), b-(iii), c-(ii), d-(i)

(3) a-(ii), b-(iii), c-(iv), d-(i)

(4) a-(iii), b-(ii), c-(iv), d-(i)

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61. Consider the following set of elements
(a) Li, Na and K
(b) Cu, Ag and Au
(c) Ca, Sr and Ba
(d) Cl, Br and I
The sets which represents Dobereiner's Triads is

(1) (a), (b) and (c) only

(2) (b), (c) and (d) only

(3) (a), (c) and (d) only

(4) (a), (b), (c) and (d)

62. Which of the following statements are incorrect?
a. The electron gain enthalpy of F is more negative than that of Cl.
b. Generally ionization enthalpy decreases down the group in periodic table.
c. The electronegativity of an atom depends upon the atoms bonded to it.
d. Cr_2O_3 and CO are examples of amphoteric oxides.
Choose the most appropriate answer from the option given below.

(1) a and d only

(2) b and d only

(3) a, b and d only

(4) a, b, c and d only

63. In which of the following, the order of arrangement agrees with the variation of property indicated against it?

(1) Na < Li < K < Rb (Metallic character)

(2) Be < B < C < N (Ionisation enthalpy)

(3) I < Br < F < Cl (Electron affinity)

(4) P < Cl < S < Si (Atomic size)

64. The correct order of ionic radii is

(1) $P^{3-} > Cl^- > S^{2-} > K^+$

(2) $K^+ > Cl^- > S^{2-} > P^{3-}$

(3) $P^{3-} > S^{2-} > Cl^- > K^+$

(4) $S^{2-} > Cl^- > K^+ > P^{3-}$

65. Given below are two statements
Statement I: Electron gain enthalpy of bromine is more negative than fluorine.
Statement II: Silicon is more electronegative than carbon as per Pauling scale.
In the light of above statements, choose the correct option.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are incorrect

(4) Both statement I and statement II are correct

66. Match list-I with list-II

	List-I Atomic number		**List-II IUPAC official name**
(a)	101	(i)	Lawrencium
(b)	103	(ii)	Mendelevium
(c)	105	(iii)	Dubnium
(d)	107	(iv)	Bohrium

Choose the correct option.

(1) (a)–(ii), (b)–(i), (c)–(iii), (d)–(iv)

(2) (a)–(ii), (b)–(iv), (c)–(i), (d)–(iii)

(3) (a)–(ii), (b)–(i), (c)–(iv), (d)–(iii)

(4) (a)–(i), (b)–(iii), (c)–(iv), (d)–(ii)

67. Match list I with list II

Column I (Compounds)	**Column II (Nature)**
a. Na_2O	(i) Acidic
b. Cl_2O_7	(ii) Amphoteric
c. As_2O_3	(iii) Neutral
d. N_2O	(iv) Basic

Choose the correct option.

(1) a(iv), b(i), c(ii), d(iii)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(ii), b(ii), c(iii), d(i)

(4) a(iii), b(ii), c(i), d(iv)

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68. Consider the following statements
(a) Gallium is called Eka-aluminium
(b) Sodium is more electropositive than potassium
(c) Electron gain enthalpy of helium is positive
The correct statements are

(1) (a) and (b) only

(2) (a) and (c) only

(3) (b) and (c) only

(4) (a), (b) and (c)

69. An electron is moving with a velocity 2×10^3 ms^{-1} with 0.01% of uncertainty. The uncertainty in the measurement of its position will approximately be ($h = 6.6 \times 10^{-34}$ Js & $m_e = 9 \times 10^{-31}$ kg)

(1) 9.1×10^{-4} m

(2) 3.0×10^{-6} m

(3) 3.0×10^{-4} m

(4) 9.1×10^{-6} m

70. The ratio of radius of 2nd Bohr orbit of Li^{2+} to the radius of first Bohr orbit of He^{+} is

(1) 4 : 3

(2) 2 : 4

(3) 8 : 3

(4) 8 : 5

71. The ratio of the wavelength of the second line of Balmer series to the third line of Balmer series of H-atom is

(1) 21 : 25

(2) 28 : 25

(3) 25 : 28

(4) 21 : 17

72. Maximum kinetic energy of photoelectrons emitted when a light of frequency 1.1×10^{12} Hz is irradiated on a metal surface whose threshold frequency is equal to 1.0×10^{11} Hz, is ($h = 6.6 \times 10^{-34}$ J s)

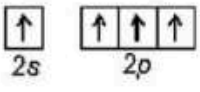
(1) 6.6×10^{-25} J

(2) 1.8×10^{-21} J

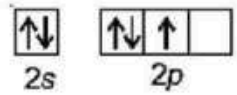
(3) 5.1×10^{-20} J

(4) 6.6×10^{-22} J

73. Which of the following orbital diagrams violates Aufbau principle?

(1) 

(2) 

(3) 

(4) 

74. Match List-I with List-II

List-I Subshells		List-II Radial Nodes
a. 4*p*	(i)	1
b. 5*s*	(ii)	2
c. 4*d*	(iii)	Zero
d. 4*f*	(iv)	4

The correct match is

(1) a(ii), b(iv), c(iii), d(i)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(iv), b(ii), c(i), d(iii)

(4) a(ii), b(iv), c(i), d(iii)

75. The maximum number of electrons in an atom that can have quantum number values n = 3, l = 1 are

(1) 6

(2) 2

(3) 10

(4) 18

76. Number of degenerate orbitals in 3rd excited state of hydrogen atom is

(1) 9

(2) 4

(3) 3

(4) 16

77. Consider the following sets of quantum numbers

	n	*l*	*m*	*s*
(a)	4	2	−2	$+\frac{1}{2}$
(b)	3	1	+2	$-\frac{1}{2}$
(c)	5	3	+1	$+\frac{1}{2}$
(d)	2	0	−1	$-\frac{1}{2}$

The correct sets are

(1) (a) and (c) only

(2) (a) and (b) only

(3) (a), (b) and (c) only

(4) (a), (b), (c) and (d)

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78. Given below are the two statements

Statement I: Magnetic quantum number gives information about the spatial orientation of the orbital with respect to standard set of co-ordinate axes.

Statement II: For $l = 0$, the only permitted value of $m_l = 0$

In light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are correct

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

79. For multielectronic species the correct order of energy of the given orbitals will be

(1) 4*f* > 4*d* > 5*p* > 5*s*

(2) 5*p* > 5*s* > 4*f* > 4*d*

(3) 4*f* > 5*p* > 4*d* > 5*s*

(4) 5*p* > 4*d* > 4*f* > 5*s*

80. Consider the following statements about cathode rays
(a) These rays start from cathode and move towards the anode.
(b) In the absence of electrical or magnetic field, these rays travel in straight lines.
(c) The Characteristics of cathode rays depend upon the material of electrodes.
In light of above statements, choose the correct answer.

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a), (b) and (c)

(4) (a) and (c) only

81. Consider the following statements
(a) AgCl has more ionic character as compared to NaCl
(b) Both SO_2 and H_2O have bent shape
(c) Dipole moment of HI is less than that of HCl
The correct statements are.

(1) a and b only

(2) a and c only

(3) b and c only

(4) a, b and c

82. During the conversion of O_2 to O_2^+ ion, the electron is removed from which of the following molecular orbitals?

(1) π^*

(2) π

(3) σ

(4) σ^*

83. Match List I with List II

	List-I Molecule		List-II Shape
a.	XeO_3	(i)	Tetrahedral
b.	$SiCl_4$	(ii)	T-Shape
c.	ClF_3	(iii)	Square planar
d.	XeF_4	(iv)	Trigonal pyramidal

The correct match is

(1) a(ii), b(iii), c(iv), d(i)

(2) a(ii), b(iv), c(iii), d(i)

(3) a(iv), b(i), c(iii), d(ii)

(4) a(iv), b(i), c(ii), d(iii)

84. Given below are the two statements one is labelled as assertion (A) the other is labelled as reason (R).
Assertion (A): Boiling point of HF is higher than HCl.
Reason (R): HF forms intermolecular hydrogen bond with each other.
In the light of above statements, choose the **correct** answer.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true, (R) is the correct explanation of (A)

85. The pair of paramagnetic species is

(1) N_2 and H_2

(2) C_2 and B_2

(3) B_2 and O_2

(4) F_2 and O_2

86. Decreasing order of stability of N_2, N_2^+ and N_2^- is

(1) $N_2^+ > N_2^- > N_2$

(2) $N_2 > N_2^- > N_2^+$

(3) $N_2 > N_2^+ > N_2^-$

(4) $N_2^- > N_2^+ > N_2$

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87. Given below are the two statements.
Statement I: A polyatomic molecule can have zero dipole moment.
Statement II: It is necessary for a polyatomic molecule to have linear geometry to have dipole moment zero
In light of above statements, choose the correct answer from the options given below.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

88. Match species given in List-I with their bond angles in List-II

	List I		List-II
a.	CH_4	(i)	107°
b.	NH_3	(ii)	109.5°
c.	BF_3	(iii)	104.5°
d.	H_2O	(iv)	120°

Choose the correct option

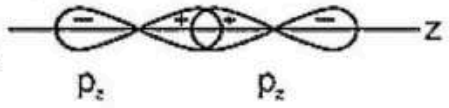
(1) a(ii), b(i), c(iv), d(iii)

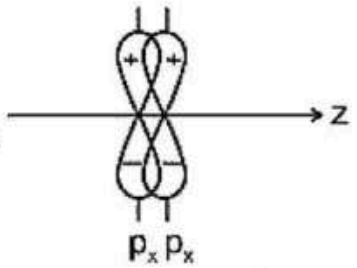
(2) a(i), b(ii), c(iii), d(iv)

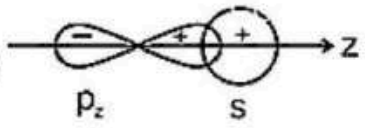
(3) a(ii), b(iv), c(i), d(iii)

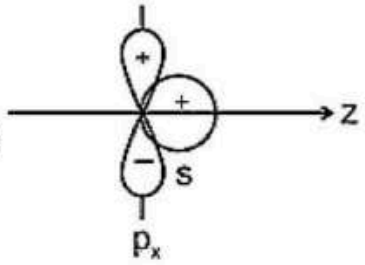
(4) a(iv), b(ii), c(iii), d(i)

89. Which of the given overlapping of orbitals leads to zero overlap?

(1) 

(2) 

(3) 

(4) 

90. Which of the following species has permanent dipole moment?

(1) BF_3

(2) XeF_4

(3) PCl_5

(4) ClF_3

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BOTANY

91. Which of the following is the most dramatic period of the cell cycle, involving major reorganisation of virtually all components of the cell?

(1) G_1 phase

(2) G_2 phase

(3) M phase

(4) Interphase

92. Select the common feature between anaphase-I and mitotic anaphase.

(1) Separation of homologous chromosomes

(2) Splitting of centromere and separation of chromatids

(3) Movement of chromosomes towards opposite poles

(4) Decondensation of chromosomes into chromatin

93. Given below are two statements:
Statement A: The compaction of chromosomes continues throughout leptotene.
Statement B: In oocytes of some vertebrates, diplotene can last for months or years.
In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both statements A and B are true

(2) Both statements A and B are false

(3) Statement A is true but statement B is false

(4) Statement A is false but statement B is true

94. In S phase of cell cycle

(1) The amount of DNA doubles

(2) Number of chromosomes doubles

(3) Number of chromosomes becomes half

(4) Organelles like mitochondria and chloroplasts duplicate

95. Which of the following events occurs during the telophase of mitosis?

(1) Chromosomal condensation

(2) Reformation of nuclear envelope

(3) Formation of spindle fibres

(4) Separation of chromatids

96. During the "Pachytene" stage of meiosis

(1) Chromosomes start pairing and form bivalents

(2) There is formation of synaptonemal complex

(3) Crossing over between sister chromatids of homologous chromosomes occurs

(4) Four chromatids of each bivalent chromosomes become distinct and clearly appear as a tetrad.

97. When karyokinesis is **not** followed by cytokinesis, it results in the

(1) Entrance of the cell into G_0 phase

(2) Formation of cell plate

(3) Haploid stage of the cell

(4) Formation of syncytium

98. What will be the amount of DNA in meiotic II products if a meiocyte at G_1 stage possesses 80 pg DNA?

(1) 20 pg

(2) 40 pg

(3) 80 pg

(4) 10 pg

99. Choose the **correct** sequence of events in prophase I.

(1) Leptotene → Zygotene → Pachytene → Diplotene → Diakinesis

(2) Leptotene → Pachytene → Zygotene → Diplotene → Diakinesis

(3) Leptotene → Zygotene → Diplotene → Pachytene → Diakinesis

(4) Leptotene → Diplotene → Zygotene → Pachytene → Diakinesis

100. Given below are two statements, one is assertion (A) and the other is reason (R).
Assertion (A): In the second phase of karyokinesis of M phase, the chromosome are spread through the cytoplasm of the cell.
Reason (R) : The complete disintegration of the nuclear envelope marks the start of the second phase of mitosis.
In the light of above statements, select the most appropriate option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Only (A) is true

(4) Both (A) and (R) are false

101. State true (T) or false (F) for the given statements and choose the correct option.
A. Meiosis increases the genetic variability
B. Meiosis maintains the chromosome number from generation to generation
C. Mitosis maintains the size of the cell

(1) A-(T), B-(T), C-(F)

(2) A-(T), B-(T), C-(T)

(3) A-(F), B-(F), C-(T)

(4) A-(T), B-(F), C-(T)

102. Match the column I with column II

Column I (Sub phases of prophase I)	**Column II (Specific characters)**
a. Zygotene	(i) Dissolution of synaptonemal complex
b. Pachytene	(ii) Synapsis between homologous chromosomes
c. Diplotene	(iii) Disappearance of nucleolus
d. Diakinesis	(iv) Exchange of genetic material between homologous chromosomes

Choose the **correct** answer from the options given below

(1) a(iv), b(ii), c(iii), d(i)

(2) a(i), b(ii), c(iv), d(iii)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iv), b(iii), c(ii), d(i)

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103. Which of the following characteristics of living beings is **not** said to be true for worker bees?

(1) Anabolism

(2) Reproduction

(3) Consciousness

(4) Growth

104. Choose **odd** one out w.r.t. class

(1) Insecta

(2) Dicotyledonae

(3) Mammalia

(4) Angiospermae

105. Read the following statements and select the **correct** option

Statement A : Families are characterised on the basis of both vegetative and reproductive features of plant species.

Statement B : Family has a group of related genera with still more number of similarities as compared to genus and species.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

106. In the taxonomic categories which hierarchical arrangement in descending order is **correct** in case of animals?

(1) Kingdom, Order, Class, Phylum, Family, Genus.

(2) Kingdom, Order, Phylum, Class, Family, Genus, Species

(3) Kingdom, Phylum, Class, Order, Family, Genus, Species

(4) Kingdom, Class, Phylum, Family, Order, Genus, Species

107. Small bristle-like fibres sprouting out in some bacteria which help them in attaching to rocks or host tissue is called

(1) Fimbriae

(2) Pili

(3) Flagella

(4) Cilia

108. The membrane of erythrocyte of human beings is composed of

(1) 50 percent protein and 22 percent lipids

(2) 40 percent protein and 52 percent lipids

(3) 42 percent protein and 50 percent lipids

(4) 52 percent protein and 40 percent lipids

109. Who gave a final shape to the cell-theory and explained that new cells are formed from pre-existing cells?

(1) Schleiden and Schwann

(2) Robert Brown

(3) Rudolf Virchow

(4) Singer and Nicolson

110. Which of the organelle has an organisation like the cartwheel?

(1) Chloroplast

(2) Ribosome

(3) Vacuole

(4) Centriole

111. Proteins modified by golgi apparatus are released from its

(1) Cis face

(2) Convex face

(3) Forming face

(4) Concave face

112. The cell organelle or structure common in both animal and higher plant cell is

(1) Centriole

(2) Large central vacuole

(3) Ribosome

(4) Plastid

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113. Main arena of cellular activities in both prokaryotes and eukaryotes is

(1) Plasma membrane

(2) Cytoplasm

(3) Nucleus

(4) Cell wall

114. Middle lamella between the plant cells is chiefly made up of

(1) Chitin

(2) Calcium pectate

(3) Cellulose

(4) Peptidoglycan

115. Bacterial viruses

a. Infect the sole members of kingdom Monera
b. Usually have double stranded DNA
c. Are also called bacteriophage
d. Generally lack protein coat

Choose the **correct** option.

(1) Only a is correct

(2) Only a and c are correct

(3) Only b is correct

(4) Only d is incorrect

116. Which of the following statements is **not** true w.r.t. the powerhouses of the cell?

(1) They are the sites of aerobic respiration

(2) Matrix possesses a circular DNA, a few RNA molecules and 40S as smaller subunit of ribosomes

(3) They divide by fission

(4) Their number depends upon the physiological activity of the cell

117. The function **not** associated with endomembrane system is

(1) Packaging of material to be delivered intra-cellular targets or out-side the cell

(2) Digestion of carbohydrates, proteins, lipids and nucleic acids

(3) Synthesis of glycolipids

(4) Maintenance of cell shape

118. Read the following statements and select the **correct** option.

Statement A : Peripheral proteins of plasma membrane are partially or totally buried in the membrane.

Statement B : Neutral solutes may move across the plasma membrane by the process of simple diffusion along the concentration gradient.

(1) Only A is incorrect

(2) Only B is incorrect

(3) Both A and B are correct

(4) Both A and B are incorrect

119. How many of the following statements is/are **correct**?

(a) The cell wall of a young plant cell is capable of growth.

(b) Secondary cell wall holds or glues the different neighbouring cells together.

(c) Middle lamella is formed on the inner side of the plant cell.

(d) The cell wall and middle lamella may be traversed by plasmodesmata which connect the cytoplasm of neighbouring cells.

(1) One

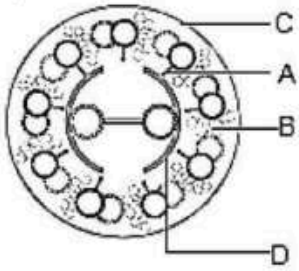
(2) Three

(3) Four

(4) Two

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120. Observe the diagram given below and choose the **correct** option.



(1) A – Radial spoke, B – Intradoublet bridge, C – Central sheath, D – Plasma membrane

(2) A – Central spoke, B – Intradoublet bridge, C – Central Microtubule, D – Plasma membrane

(3) A – Radial spoke, B – Interdoublet bridge, C – Central sheath, D – Central Microtubule

(4) A – Radial spoke, B – Interdoublet bridge, C – Plasma membrane, D – Central sheath

121. Match the following columns and select the **correct** option.

Column-I	Column-II
a. Elaioplast	(i) Store proteins
b. Aleuroplast	(ii) Colourless plastids
c. Chromoplast	(iii) Store fats and oils
d. Leucoplast	(iv) Carotenoid pigments

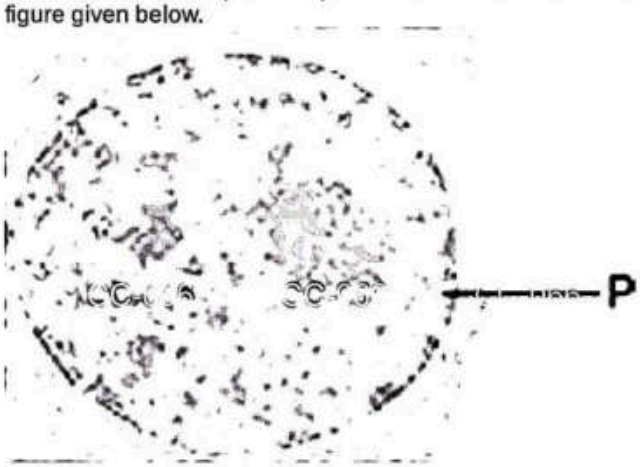
(1) a(iv), b(iii), c(i), d(ii)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(ii), b(iv), c(iii), d(i)

(4) a(i), b(ii), c(iv), d(iii)

122. Select the **correct** option for part 'P' of the structure in the figure given below.



(1) Nuclear pore – Forms a barrier between the materials present inside the nucleus and that of the cytoplasm

(2) Nucleoplasm – Contains nucleolus and chromatin

(3) Nucleoplasm – Constitutes the nuclear matrix

(4) Nuclear pore – Passage through which movement of RNA and protein molecules takes place in both directions between nucleus and cytoplasm

123. All are correct statements w.r.t. phycomycetes, **except**

(1) Aplanospores are exogenously produced

(2) Aseptate and multinucleated mycelium

(3) They can be aquatic in habitat

(4) Formation of zygospores

124. Select the **correct** option to fill the blanks.

Characters	Monera	Protista	Fungi
Cell type	A	Eukaryotic	B
Cell wall	Non-cellulosic	C	Chitin
Body organisation	D	Cellular	E

(1) A-Prokaryotic, B-Eukaryotic, D-Cellular

(2) B-Eukaryotic, D-Tissues, E-Organ

(3) A-Eukaryotic, C-Absent, E-Loose tissue

(4) B-Prokaryotic, C-Chitin, D-Cellular

125. Archaebacteria that often inhabit hot spring are

(1) Thermoacidophiles

(2) *Methanococcus*

(3) Halophiles

(4) Methanogens

126. Heterocysts present in *Anabaena* are specialised for

(1) Nitrogen fixation

(2) Food storage

(3) Fission

(4) Sexual reproduction

127. State true (T) or false (F) for the given statements and select the **correct** option.
(A) Fungi are heterotrophic and absorb soluble organic matter from dead substrate.
(B) Bladderwort and Venus fly trap are insectivorous plants and *Cuscuta* is a parasite.
(C) In *Agaricus* and *Puccinia*, mycelium is generally aseptate and coenocytic.
(D) Fusion of protoplasm between two motile or non-motile gametes is called plasmogamy.

(1) (A)T, (B)T, (C)F, (D)T

(2) (A)F, (B)F, (C)T, (D)F

(3) (A)T, (B)F, (C)T, (D)F

(4) (A)F, (B)T, (C)F, (D)T

128. All of the following are heterotrophic protists, **except**

(1) Slime moulds

(2) Diatoms

(3) *Paramoecium*

(4) Sporozoans

129. Mycelium as continuous tubes filled with multinucleated cytoplasm is found in

(1) *Saccharomyces* and *Penicillium*

(2) *Rhizopus*, *Mucor* and *Albugo*

(3) *Rhizopus*, *Ustilago* and *Puccinia*

(4) *Alternaria*, *Colletotrichum* and *Albugo*

130. The vast majority of bacteria

(1) Can synthesize their own food from inorganic substances

(2) Use sunlight to synthesize their food

(3) Are heterotrophic in nature

(4) Are complex in structure but simple in behaviour

131. The causal agent of bovine spongiform encephalopathy

(1) Has RNA as genetic material

(2) Is similar in structure and function to viruses

(3) Does not contain any nucleic acid

(4) Is enveloped with phospholipid membrane

132. Identify the **incorrect** statement w.r.t. dinoflagellates.

(1) These organisms are mostly marine and heterotrophic

(2) The cell wall has stiff cellulosic plates on the outer surface

(3) Most of them have two flagella

(4) *Gonyaulax* is responsible for red tides

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133. Which of the following features is common amongst *Alternaria*, *Agaricus*, *Aspergillus* and *Rhizopus*?

(1) Their mycelia are branched and septate

(2) They sexually reproduce by forming non-motile gametes

(3) They asexually reproduce by the formation of planospores

(4) These are terrestrial fungi

134. Dikaryophase is seen

(1) Only in basidiomycetes

(2) When plasmogamy is not immediately followed by karyogamy

(3) Only in those fungi where sexual spores are formed endogenously

(4) In imperfect fungi

135. Read the following features.
(a) Help in mineral cycling.
(b) Lack karyogamy and meiosis
(c) Conidia are asexual spores.
Above given features are **true** for the member of group

(1) Ascomycetes

(2) Basidiomycetes

(3) Phycomycetes

(4) Deuteromycetes

ZOOLOGY

136. Choose the incorrect option w.r.t. connective tissues.

(1) Presence of modified polysaccharides that act as the ground substance

(2) Composed of matrix, fibres and cells

(3) Forms packing around the organs for protection

(4) Have fibres that provide elasticity, excitability, strength and flexibility

137. Which of the following junctions exist in our skin that prevents the entry of water into body?

(1) Gap junctions

(2) Tight junctions

(3) Hemidesmosomes

(4) Desmosomes

138. Which of the given features is **not** related to cardiac muscle fibres?

(1) Involuntary and autorhythmic

(2) Branched and striated

(3) Cylindrical and have intercalated discs

(4) Structural syncytium and have communication junctions

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139. Identify the structures which are **not** available in all connective tissue
a. Fibroblasts
b. Fibres
c. Macrophages
d. Matrix

(1) a and c only

(2) a and b only

(3) a , b and c only

(4) a, b and d only

140. Bone differs from cartilage in possessing

(1) Solid ground substance

(2) Cells in fluid filled spaces

(3) Matrix which is rich in calcium salts and collagen fibres

(4) Intercellular material that resists compression

141. **Assertion (A):** In a cardiac muscle, when one cell receives a signal to contract, its neighbours are also stimulated to contract.
Reason (R): Communication junctions at some fusion points allow the cells to contract as a unit.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

142. Consider the given features :
(a) Present beneath the skin
(b) Excess of nutrients which are not used immediately are converted into fats and stored in this tissue
The above given features are true for the tissue that belongs to the category of

(1) Dense regular connective tissue

(2) Loose connective tissue

(3) Dense irregular connective tissue

(4) Fluid connective tissue

143. In humans, the type of epithelium present in the inner lining of pharynx is also found in

(1) Moist surface of buccal cavity

(2) Inner lining of stomach

(3) Inner lining of aorta

(4) Lining of air sacs of lungs

144. Consider the given features.
(a) Possesses compactly packed fibres and fibroblasts
(b) Attaches skeletal muscles to bones
(c) Orientation of fibres show a regular pattern
Choose the option for which all of the above given features hold true.

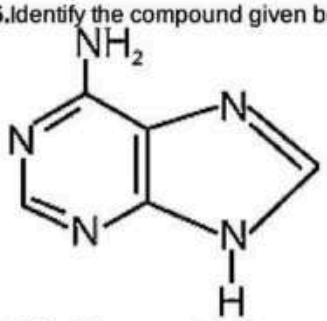
(1) Tendon

(2) Ligament

(3) Adipose tissue

(4) Areolar tissue

145. Identify the compound given below



Select the **correct** option.

(1) Adenine

(2) Guanine

(3) Thymine

(4) Cytosine

146. How many among the following are secondary metabolite(s)?

Morphine, Insulin, Vinblastin, Cellulose, Glucose, Sucrose, Ricin, Anthocyanin

(1) 5

(2) 4

(3) 3

(4) 8

147. Which one of the following enzymes contains 'Haem' as the prosthetic group?

(1) Succinate dehydrogenase

(2) Catalase

(3) RuBisCo

(4) Carbonic anhydrase

148. Which of the following statements is **incorrect** w.r.t. uracil?

(1) It is a heterocyclic compound

(2) It is present in RNA

(3) It forms uridylic acid when it combines with deoxyribose sugar and phosphate

(4) It forms glycosidic bond with sugar

149. Read the following statements and choose the **correct** option.
Statement I : In solutions of different pH, the structure of amino acids changes.
Statement II : An amino acid may exist as zwitterionic form, anionic form or cationic form depending on the pH of the solution.

(1) Only statement I is correct

(2) Only statement II is correct

(3) Both the statements I and II are correct

(4) Both the statements I and II are incorrect

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150. The EC number of an enzyme that catalyses the formation of carbonic acid from CO_2 and H_2O is 4.2.1.1. This enzyme belongs to the class named

(1) Dehydrogenases

(2) Hydrolases

(3) Isomerases

(4) Lyases

151. Choose the aromatic amino acid among the following which must be obtained through diet as humans are incapable of synthesising it.

(1) Cysteine

(2) Tryptophan

(3) Proline

(4) Alanine

152. Major difference between enzyme and inorganic catalysts is

(1) Former works efficiently on all range of temperatures

(2) Latter works efficiently at high temperature and high pressure

(3) Former works only at 95°C and above that it gets inactive

(4) Latter is specific about temperature and would not work above 40°C

153. The molecular weights of nucleic acids fall in the range of _____ and above, while the molecular weight of lipids is _____. Choose the option that correctly fills the respective blanks.

(1) 10,000 Da; more than 5000 Da

(2) 10,000 Da; twice 1000 Da

(3) 10,000 Da; less than 800 Da

(4) 100 Da; more than 10,000 Da

154. Select the option which is **mismatched**.

(1)	Transferases	–	Transfer of specific groups other than hydrogen from one substrate to another
(2)	Isomerases	–	Catalyse the rearrangement of molecular structure to form isomers
(3)	Ligases	–	Catalyse the cleavage of substrate into two parts, without hydrolysis
(4)	Dehydrogenases	–	Catalyse oxidation-reduction reaction involving transfer of electrons

(1) (1)

(2) (2)

(3) (3)

(4) (4)

155. Which of the following factor(s) do(es) not affect enzyme activity?
A. Temperature
B. pH
C. Enzyme concentration
D. Product concentration
E. Substrate concentration
F. Activation energy

(1) C only

(2) C and D

(3) D only

(4) F only

156. **Assertion (A)** : K_m value is the concentration of substrate at which half of V_{max} is achieved.
Reason (R) : K_m value decreases with increase in amount of competitive inhibitor.
In the light of above statements, choose the correct answer.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

157. Comprehend the given statements
a. Collagen forms intracellular ground substance.
b. Adult human haemoglobin consists of 4 subunits and all of them are identical to each other.
c. Competitive inhibitors are often used for the control of bacterial pathogens.
Which of the above statements is/are **correct**?

(1) a and c only

(2) Only c

(3) a and b only

(4) b and c only

158. A dehydration reaction links two glucose molecules to produce maltose. If the formula for glucose is $C_6H_{12}O_6$, then what is the formula of maltose?

(1) $C_{12}H_{22}O_{11}$

(2) $C_{12}H_{24}O_{11}$

(3) $C_{12}H_{20}O_{10}$

(4) $C_{12}H_{24}O_{12}$

159. All of the following bonds are likely to be involved in stabilizing three-dimensional folding of most proteins **except**

(1) Hydrogen bonds

(2) Covalent bonds

(3) Ionic bonds

(4) Phosphodiester bonds

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160. The volume of air that will remain in the lungs after a normal expiration is equivalent to

(1) TLC

(2) ERV + RV

(3) TV + ERV

(4) IC + ERV

161. Lungs do not collapse in between breaths because some air always remains in the lungs which is called

(1) Inspiratory reserve volume

(2) Expiratory capacity

(3) Expiratory reserve volume

(4) Residual volume

162. Under normal physiological condition, during the formation of oxyhaemoglobin, about 90% saturation of Hb with O_2 occurs when the pO_2 is about

(1) 40 mm Hg

(2) 60 mm Hg

(3) 55 mm Hg

(4) 45 mm Hg

163. Amongst all the given respiratory surfaces, which of the following is/are associated with capillary beds?
(a) Alveoli of lungs in humans
(b) Skin of frogs
(c) Cuticle of earthworms
Choose the **correct** option.

(1) (a) only

(2) (a) and (b) only

(3) (b) and (c) only

(4) (a), (b) and (c)

164. Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli under normal physiological conditions.

(1) Low pO_2, low pCO_2, more H^+, higher temperature

(2) High pO_2, low pCO_2, less H^+, lower temperature

(3) Low pO_2, high pCO_2, more H^+, higher temperature

(4) High pO_2, high pCO_2, less H^+, higher temperature

165. Abdominal muscles directly play a role in breathing mainly during

(1) Normal inspiration

(2) Normal expiration

(3) Forceful expiration

(4) Forceful inspiration

166. The partial pressure of oxygen in the systemic arteries is 95 mm Hg which is

(1) Less than the pO_2 in lung alveoli

(2) Less than the pCO_2 in systemic veins

(3) Equal to the pCO_2 in systemic veins

(4) Less than the pO_2 in actively metabolising tissues

167. Enzyme that facilitates the formation of carbonic acid at tissue level and is present in high concentration within RBCs, is

(1) Carbonic anhydrase

(2) Carboxypeptidase

(3) Peroxidase

(4) Catalase

168. Under normal physiological conditions, a gas 'X' does not play any significant role in regulation of breathing, and its partial pressure in deoxygenated blood is 40 mm Hg. Select the **correct** statement w.r.t. 'X'.

(1) 100 mL of oxygenated blood can deliver 5 mL of 'X' to tissues.

(2) 100 mL of oxygenated blood can deliver 4 mL of 'X' to alveoli.

(3) Changes in concentration of 'X' are only recognised by aortic arch.

(4) Only 80% of 'X' is transported by RBCs.

169. In humans, the specialized centre that can moderate the functions of respiratory rhythm centre (present in the medulla oblongata) is named 'X' . Choose the correct option w.r.t 'X'.

(1) Recognises changes in O_2 and H^+ concentration and send necessary signals to the rhythm centre for remedial actions

(2) Can reduce the duration of inspiration and thereby alter the respiratory rate

(3) Situated adjacent to chemosensitive area which is highly sensitive to hydrogen ions only

(4) When 'X' is active, breathing becomes shallower and slower

170. During a normal cardiac cycle in humans, just before the beginning of atrial contraction, ventricles are

(1) Nearly two third filled by blood

(2) Only one-third filled by blood

(3) Completely filled by blood

(4) Empty

171. A person can be diagnosed with hypertension, if on repeated checkups, his

(1) Systolic pressure is below 80 mm Hg

(2) Systolic pressure is below 120 mm Hg

(3) Systolic and diastolic pressures are beyond 140 and 90 mm Hg respectively on repeated check up

(4) Systolic and diastolic pressures are below 80 and 120 mm Hg respectively

172. Choose the correct statement w.r.t the cardiac cycle of a healthy individual under normal physiological conditions.

(1) The volume of blood pumped out by the right and left ventricles is same.

(2) The volume of blood pumped out by the right and left ventricles is significantly different.

(3) Complete filling of ventricles occur during joint diastole.

(4) Amount of blood pumped out in each cardiac cycle is called cardiac output.

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173. Read the following statements carefully
(a) Interstitial fluid has the same mineral distribution as that of plasma.
(b) Tissue fluid helps in exchange of gases.
(c) Fats are absorbed through blood in lacteals.
Choose the correct option.

(1) Only statement (c) is correct

(2) Only statements (a) and (b) are correct

(3) All statements (a), (b) and (c) are correct

(4) Only statements (b) and (c) are correct

174. Match column I with column II and choose the correct option w.r.t a standard ECG of a healthy man.

	Column I		Column II
a.	P-wave	(i)	Represents the repolarization of ventricles
b.	T-wave	(ii)	Represents the electrical excitation of the atria
c.	Q-wave	(iii)	Helps to determine the heart beat rate
d.	QRS-complex	(iv)	Marks the beginning of the ventricular systole

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(iv), b(iii), c(ii), d(i)

175. Read the following statements:
(a) All chordates have a closed circulatory system.
(b) All vertebrates are characterised by presence of a closed circulatory system.
(c) In fishes, heart pumps out oxygenated blood.
Choose the **correct** option.

(1) (a), (b) and (c) are correct

(2) Only (a) and (b) are correct

(3) Only (a) and (c) are correct

(4) Only (b) is correct

176. Comprehend the given statements about circulatory system of humans.
(a) Damage to chordae tendinae of the tricuspid valves will immediately decrease the blood flow to aorta.
(b) The opening between left atrium and left ventricle is guarded by bicuspid valve.
Choose the **correct** option.

(1) Both statements (a) and (b) are correct

(2) Statement (a) is correct but statement (b) is incorrect

(3) Statement (a) is incorrect but statement (b) is correct

(4) Both statements (a) and (b) are incorrect

177. What type of blood group can be transfused to a person who has anti-B antibodies in his plasma but no Rh antigens on the surface of his RBCs?

(1) AB^{+ve}

(2) B^{+ve}

(3) A^{-ve}

(4) O^{+ve}

178. Proteins contribute 'X' % of the plasma in human blood. The value of 'X' is equal to the percentage of 'Y' among total WBCs. Choose the correct function of 'Y'.

(1) Secrete histamine, serotonin and heparin

(2) Mainly resist infections and are also associated with allergic reactions

(3) Phagocytic cells with bean shaped nucleus, destroy foreign organisms entering the body

(4) Granulocytes that are responsible for immune responses of the body *via* production of antibodies

179. At the beginning of ventricular systole, rise in ventricular pressure causes the AV valves to close. Further rise in ventricular pressure eventually leads to

(1) Opening of AV valves

(2) Atrial depolarisation

(3) Opening of semilunar valves

(4) Rapid filling of left atrium

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180. Match column I and column II w.r.t epithelium and its location.

	Column I		Column II
(i)	Ciliated columnar	(a)	Stomach
(ii)	Brush border columnar	(b)	PCT
(iii)	Simple columnar	(c)	Intestine
(iv)	Brush border cuboidal	(d)	Fallopian tube

Choose the **correct** match

(1) i(d), ii(a), iii(b), iv(c)

(2) i(b), ii(c), iii(a), iv(d)

(3) i(d), ii(c), iii(a), iv(b)

(4) i(b), ii(a), iii(c), iv(d)